

1. Find Drag Polar Calculations we find: Data to find Airfoil 7

$C_{L, cruise}$, $C_{L, minimum}$
 $C_{D, cruise}$, $C_{D, min}$, $C_{D,0}$
 $k_D = \frac{C_D}{C_L}$, cruise efficiency

$$C_{L, max} = \frac{2W}{\rho V_{stall}^2 S}$$

$$Re = \frac{\rho V_{cruise} \bar{c}}{\mu}$$

μ = dynamic viscosity
 - Result when using this to find airfoil
 $\frac{t}{c}$ = thickness ratio, airfoil camber, zero-lift angle, lift curve slope, & C_L vs α graph



Location of CG: $x_{CG} = \frac{\sum W_i x_i}{\sum W_i}$ - the more is $x_i = 0$ m

1. Taper Ratio for Wing: we use average chord length for calc but we then find tip & root chords for that average chord length by:

Taper ratio $\lambda = \frac{C_t}{C_r}$

choose a taper ratio that fits the type of aircraft & then
 ~~$C_t = \lambda \cdot C_r$ & $C_r = \bar{c} \cdot \frac{3(1+\lambda)}{2(1+\lambda+\lambda^2)}$~~

Notes

- wing incidence angle, the actual angle of wing being mounted on fuselage
- α_{eff} , angle of wing to airfoil & change due to nose up & down during flight

• Zero-lift angle is the cambered airfoil, upper surface more curved than lower surface so, wing creates lift even though parallel to the airflow

Calculate empty ^{mom} ~~bold~~ forward & aft CG *

- Find neutral point, static margin - longitudinal trim
- CG range - control/elevator authority
- Performance verification of stall & cruise speed, rate of climb, service ceiling, takeoff distance, landing distance, range & endurance (basically go through ARD again)
- Limit load factor
- Ultimate load factor
- Wing bending loads
- Spar struts
- Landing gear loads

1. Neutral point location expressed as a percentage of the MAC

~~$$x_{np} = \frac{x_{ac,w}}{\bar{c}} + \eta_h \cdot V_H \left(\frac{a_h}{a_w} \right) \left(1 - \frac{dE}{d\alpha} \right)$$~~

- $x_{ac,w}$ = wing aeroc. center location
- S_h = horizontal stabilizer wing area
- S_w = wing area
- a_w = wing lift-curve slope
- a_h = tail lift-curve slope
- η_h = tail efficiency
- $\frac{dE}{d\alpha}$ = downwash gradient
- V_H = horizontal tail volume coefficient